

Gut Health and Arthritis: The Microbiome Connection Explained

Explore the link between gut health and arthritis. How your microbiome affects joint inflammation, best foods for gut health and practical steps to improve your gut.

By **Living With Arthritis Editorial Team** · Reviewed by **Clinical Advisory Panel** (HCPC-registered physiotherapy & rheumatology nursing input) · 2026-01-08 · livingwitharthritis.org.uk



*Gut-healthy fermented foods including yoghurt, kefir and berries for arthritis inflammation management
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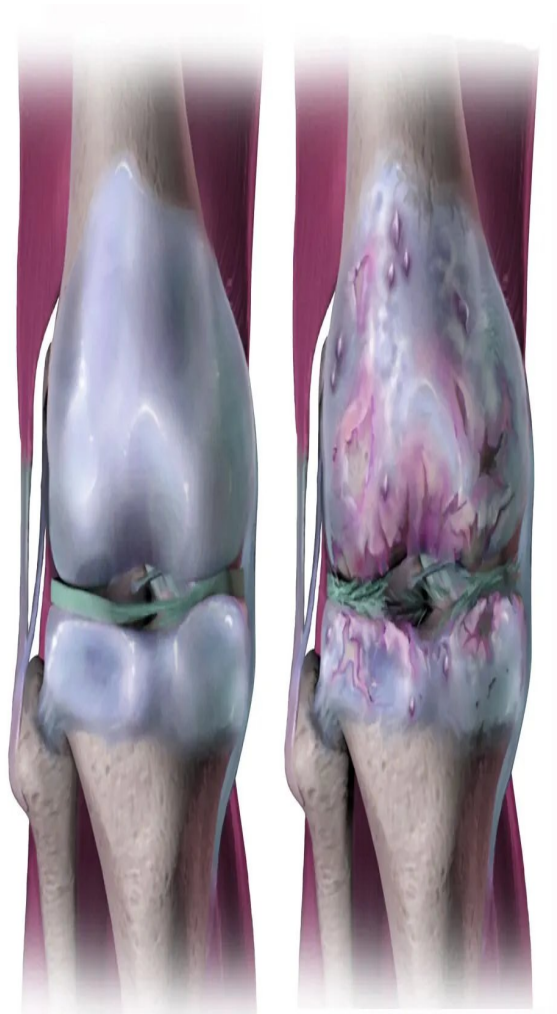
One of the most exciting areas of arthritis research in recent years has been the discovery that the **trillions of bacteria living in your gut** — collectively known as the

gut microbiome — play a significant role in inflammation and immune function throughout your entire body, including your joints. This emerging science is reshaping our understanding of why arthritis develops and how dietary changes might help manage it. This guide explains the gut-arthritis connection in plain English and provides practical steps you can take today.

What Is the Gut Microbiome?

Your gut contains approximately **100 trillion microorganisms** — bacteria, fungi, viruses and other microbes — collectively weighing about 2 kg. This community of microbes, predominantly in the large intestine, is unique to each individual and influenced by diet, medications, lifestyle and environment. A healthy microbiome contains a diverse range of beneficial bacteria that aid digestion, produce vitamins, train the immune system and help regulate inflammation throughout the body.

How Gut Health Affects Arthritis



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The Immune System Connection

Approximately **70% of your immune system** resides in or around the gut. The gut-associated lymphoid tissue (GALT) is the largest immune organ in the body. Gut bacteria directly influence how the immune system develops and functions. In inflammatory arthritis (RA, PsA), the immune system mistakenly attacks joints — and there's growing evidence that imbalances in gut bacteria (dysbiosis) may trigger or perpetuate this autoimmune response.

Intestinal Permeability ("Leaky Gut")

When the gut barrier becomes more permeable than normal, bacterial fragments and other molecules can cross into the bloodstream, triggering widespread immune activation and inflammation. Research has found increased intestinal permeability in people with rheumatoid arthritis, ankylosing spondylitis and psoriatic arthritis. While "leaky gut" is sometimes

overhyped in alternative health circles, the underlying science of intestinal permeability and its role in autoimmunity is legitimate and actively researched.

Specific Bacterial Links

- Studies have found that people with RA have different gut bacterial profiles compared to healthy individuals — notably reduced diversity and specific bacterial imbalances
- The bacterium *Prevotella copri* has been found in higher levels in people with new-onset RA
- Reduced levels of beneficial bacteria (particularly *Bifidobacterium* and *Bacteroides*) have been observed in RA patients
- The gut microbiome of people with psoriatic arthritis shows characteristic alterations that overlap with patterns seen in psoriasis and inflammatory bowel disease

Osteoarthritis Too

Even osteoarthritis — traditionally viewed as a "wear and tear" condition — has a gut connection. Research from the University of Rochester found that an unhealthy gut microbiome (induced by a high-fat diet in animal studies) accelerated OA progression. The mechanism involves systemic low-grade inflammation driven by gut bacterial products entering the bloodstream. Obesity-related OA may partly work through this gut-inflammation pathway.

Foods That Support Gut Health

Prebiotic Foods (Feed Good Bacteria)

- Vegetables — Onions, garlic, leeks, asparagus, artichokes, chicory root
- Fruits — Bananas (especially slightly underripe), apples, berries
- Whole grains — Oats, barley, rye bread
- Legumes — Lentils, chickpeas, beans
- Nuts and seeds — Almonds, flaxseeds

Probiotic Foods (Contain Live Beneficial Bacteria)

- Live yoghurt — Look for "live cultures" on the label. Available in every UK supermarket

- Kefir — A fermented milk drink increasingly available in UK shops (Biotiful, Yeo Valley)
- Sauerkraut — Unpasteurised, from the chilled section (pasteurised versions have no live bacteria)
- Kimchi — Fermented Korean vegetables, available in most UK supermarkets
- Kombucha — Fermented tea, widely available in the UK
- Miso — Fermented soybean paste, excellent in soups

Polyphenol-Rich Foods

Polyphenols are plant compounds that beneficial gut bacteria thrive on: dark chocolate (70%+ cocoa), green tea, berries, red grapes, extra-virgin olive oil, red onions and herbs/spices.

Foods That May Harm Gut Health



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- Ultra-processed foods — Emulsifiers, artificial sweeteners and additives may damage the gut barrier
- Excessive sugar — Feeds less beneficial bacteria and reduces diversity
- Excessive alcohol — Damages the gut lining and disrupts bacterial balance
- Unnecessary antibiotics — Can devastate beneficial bacteria (always take when prescribed, but don't request for viral infections)

Probiotic Supplements

The evidence for probiotic supplements in arthritis is still developing:

- Several small studies have shown modest benefits of specific probiotic strains for RA symptoms
 - A meta-analysis found probiotic supplementation reduced inflammatory markers (CRP) in RA patients
 - The most studied strains for inflammation include Lactobacillus and Bifidobacterium species
 - Evidence is promising but not yet strong enough for formal clinical recommendations
 - If you try a probiotic, choose a reputable brand with specific named strains and use for at least 4 weeks
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Practical Steps

- Eat 30 different plants per week — Diversity is the single most important factor for a healthy microbiome. Count vegetables, fruits, nuts, seeds, grains, legumes and herbs
- Include a fermented food daily — Live yoghurt, kefir, sauerkraut or kimchi
- Eat more fibre — Most UK adults eat only 18g/day; the target is 30g. Increase gradually to avoid bloating
- Reduce ultra-processed food — Cook from scratch when possible
- Stay hydrated — Water supports gut function
- Exercise regularly — Physical activity independently improves microbiome diversity

The gut-arthritis connection is a rapidly evolving field with genuine potential to improve arthritis management. While we're still learning the details, the dietary steps above will support both gut health and overall well-being. Our virtual assistant can help you plan gut-friendly meals.

This article is for general information only. The gut microbiome is a complex area of active research. Consult your GP before making major dietary changes or starting supplements.

Q&A

Question: What Is the Gut Microbiome?

Short answer: Your gut contains approximately 100 trillion microorganisms — bacteria, fungi, viruses and other microbes — collectively weighing about 2 kg. This community of microbes, predominantly in the large intestine, is unique to each individual and influenced by diet, medications, lifestyle and environment. A healthy microbiome contains a diverse range of beneficial bacteria that aid digestion, produce vitamins, train the immune system and help regulate

Question: When should I seek professional advice?

Short answer: If symptoms persist or worsen, pain is severe and unresponsive to usual relief, or you develop new symptoms, contact your GP, physiotherapist or rheumatology team promptly.

Question: Where can I get free arthritis support in the UK?

Short answer: Arthritis UK: 0800 5200 520. NRAS: 0800 298 7650. Free tools and guides at livingwitharthritis.org.uk.

General information only — not medical advice. Consult your GP or specialist before changing medication or starting a new programme. Emergency: 999. NHS advice: 111. Living With Arthritis · livingwitharthritis.org.uk